

Road traffic noise linked to immediate heart and blood vessel stress

Download PDF Copy

Interviews (/medical/interviews)



Add News Medical on Google
as a preferred source

(<https://www.google.com/preferences/source?q=news-medical.net>)

White Papers (/medical/whitepapers)

European Society of Cardiology (<https://www.escardio.org/>)

More...

Feb 25 2026

Reviewed

A new study published in the journal *Cardiovascular Research* shows that a single night of road traffic noise, at levels that are typical of those living in a city, causes stress to the heart and blood vessels.

The findings could help explain why people exposed to long-term traffic noise have higher rates of high blood pressure and heart disease.

The randomized, double-blind crossover study of 74 healthy adults found that low-level nighttime road traffic noise impaired blood vessel function and raised heart rate, as well as disrupted sleep. The study also found blood protein changes linked to inflammation and stress in participants exposed to nighttime noise.

Even a single night of road traffic noise stressed the cardiovascular system. We didn't expect to find such consistent biological changes in people exposed to noise levels typical of someone living near a road."

Dr. Omar Hahad, lead author, University Medical Center of the Johannes Gutenberg-University, Germany

The findings support calls from the European Society of Cardiology and the World Health Organization for stricter noise regulation. They underscore the need for urban planning that protects sleep and cardiovascular health in increasingly noise-polluted environments.

Seventy-four healthy participants took part for in the study for three nights. Each night they were



crossed to one of three noise conditions: no noise (control), 30 episodes of traffic noise, or 60 episodes of traffic noise. The study was double-blinded which means that neither the participants, nor the researchers, knew which sound level conditions they were given each night. About (/medical/about)

The following morning the participants had a number of health tests, including flow mediated dilation, a standard test of blood vessel function. A lower percentage dilation of the blood vessels in this test indicates impaired blood vessel function and is associated with increased risk of cardiovascular disease. The flow-mediated dilation rate for the control group was 9.35%, whereas for participants exposed 30 incidents of nighttime traffic noise it was 8.19%, and for 60 nighttime noise events it was 7.73%. News (/medical/news)

Analysis of blood samples from participants exposed to noise showed changes in interleukin signalling and chemotaxis in participants who had also experienced the worst blood vessel effects. These changes are involved in inflammation (<https://www.news-medical.net/health/What-Does-Inflammation-Do-to-the-Body.aspx>) and stress responses. Medical Devices (/Clinical-and-Diagnostics) Interviews (/medical/interviews) White Papers (/medical/whitepapers)

"These are similar key biological pathways that we find changed by noise in multiple mouse exposure studies, which means that we can now explain the molecular pathomechanisms induced by noise in humans by preclinical mechanistic insights" explained Prof. Andreas Daiber, Head of the Molecular Cardiology research group at the University Medical Center of the Johannes Gutenberg-University, and coordinator of the EU environmental research consortium MARKOPOLO. Merged (/medical/merged-a-z.aspx)

Noise exposure also increased the mean heart rate of participants by a difference of 1.23 beats per minute. Self-reported sleep quality and restfulness were significantly reduced across all dimensions after exposure to noise.

"Even when we're asleep, our bodies are still listening. Repeated activation of stress responses night after night may help explain why people exposed to long-term traffic noise have higher rates of high blood pressure and heart disease. Protecting our sleep by tackling noise pollution should be part of how we think about preventing cardiovascular disease in towns and cities around the world." Dr. Hahad continued.

Participants were required to abstain from alcohol, caffeine, nicotine, recreational substances and strenuous physical activity throughout the study. They played the sound recordings in their bedroom overnight, with compliance monitored by continuous sound level recording. Traffic noises were real-life traffic recordings and each traffic noise event reached peak sounds of approximately 60 decibels.

Dr. Hahad explained the implications of the study for individuals and policymakers, "Reducing bedroom noise exposure where possible is a reasonable measure – by moving bedrooms away from the road or buying highly insulated windows. Earplugs might also reduce the noise burden, although we don't yet have solid evidence they protect against cardiovascular risks. Making lifestyle changes such as healthier diets and physical exercise would also be protective."

Prof. Thomas Münzel, Senior author and Chair of the European Society of Cardiology Taskforce Functional Food ([/condition/Functional-Food](#)) Environment and Sustainability commented: "Transportation noise should be formally recognised in future clinical practice guidelines on cardiovascular prevention as an independent, non-traditional cardiovascular risk factor, given consistent epidemiological and mechanistic evidence linking chronic exposure to hypertension, ischemic heart disease, stroke, and cardiometabolic dysfunction at levels below current regulatory thresholds."

The researchers note that this was a short-term study in young and healthy adults, and more research is needed to understand how these results relate to long-term risk of cardiovascular disease. They would like to confirm the changes to blood proteins in a larger study in future.

According to recent European Environment Agency (EEA) estimates, approximately 150 million people over 30% of the population in the European Economic Area, which includes all European Union member states as well as Iceland, Liechtenstein and Norway, are exposed to long-term unhealthy transportation noise levels (over 55 decibels Day-Evening-Night noise levels or L_{den}) when assessed against the World Health Organization's guideline values, with road traffic being the dominant contributor, accounting for the majority of exposed individuals.

Source:

European Society of Cardiology (<https://www.escardio.org/>)

Journal reference:

Hahad, O., *et al.* (2026) A randomized, double-blind, crossover study of acute low-level night-time road traffic noise: effects on vascular function, sleep, and proteomic signatures in healthy adults. *Cardiovascular Research*. DOI: 10.1093/cvr/cvag028. <https://academic.oup.com/cardiovascres/advance-article/doi/10.1093/cvr/cvag028/8493284> (<https://academic.oup.com/cardiovascres/advance-article/doi/10.1093/cvr/cvag028/8493284>)

Be the first to rate this article



Posted in: [Medical Science News \(/category/Medical-Science-News.aspx\)](#) | [Medical Research News \(/category/Medical-Research-News.aspx\)](#) | [Medical Condition News \(/category/Medical-Condition-News.aspx\)](#)

[Comments \(0\)](#)

[Download PDF Copy](#)

Suggested Reading

News (/medical/news)

Health A-Z (/medical-a-z.aspx)

Drugs (/drugs-a-z.aspx)

[Exercise-based cardiac rehabilitation offers lasting heart health benefits](/news/20260917/Exercise-based-cardiac-rehabilitation-offers-lasting-heart-health-benefits)
(/news/20260917/Exercise-based-cardiac-rehabilitatio...

[Why frailty and heart health may be more closely connected than we thought](/news/20260915/Why-frailty-and-heart-health-may-be-more-closely-connected-than-we-thought)
(/news/20260915/Why-frailty-and-heart-health-m...

[Your voice could add something aging blood tests](/news/20260915/Your-voice-could-add-something-aging-blood-tests)
(/news/20260915/Your-voice-could-add-somethin...

[Children who ate healthier plant foods had slightly lower blood pressure later](/news/20260922/Children-who-ate-healthier-plant-foods-had-slightly-lower-blood-pressure-later)
(/news/20260922/Children-who-ate-healthier-plant...

White Papers (/medical/whitepapers)

More...

[Norwegian scientists harvest super donor cells to fight diverse diseases](/news/20260916/Norwegian-scientists-harvest-super-donor-cells-to-fight-diverse-diseases)
(/news/20260916/Norwegian-scientists-harvest-super-...

[Chronic stress and inflammation cause silent structural heart damage](/news/20260918/Chronic-stress-and-inflammation-cause-silent-structural-heart-damage)
(/news/20260918/Chronic-stress-and-inflammation-...

[Regular exercisers show distinct metabolic and immune signatures in blood](/news/20260921/Regular-exercisers-show-distinct-metabolic-and-immune-signatures-in-blood)
(/news/20260921/Regular-exercisers-show-distinct-...

[ACC Latin America 2026 studies highlight sex and regional disparities in heart attack care](/news/20260917/ACC-Latin-America-2026-studies-highlight-sex-and-regional-disparities-in-heart-attack-care)
(/news/20260917/ACC-...

Comments

The opinions expressed here are the views of the writer and do not necessarily reflect the views and opinions of News Medical.



Post a new comment

Login

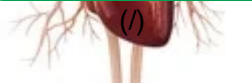
Quirky Comment Title (optional)

Post

Trending
Stories

Latest
Interviews

Top Health
Articles



(/news/20260916/Could-a-daily-glass-of-orange-juice-support-heart-health-Study-finds-citrus-compounds-influence-genes-linked-to-blood-vessel-function.aspx)



(/news/20260916/Study-defines-early-onset-cirrhosis-caused-by-fatty-liver-disease.aspx)



(/news/20260918/Salivary-cortisol-patterns-predict-cognitive-decline-in-older-adults.aspx)

(/news/20260916/POTS-identified-in-nearly-two-thirds-of-Long-COVID-patients.aspx)

(/news/20260916/Study-

Could a daily glass of orange juice support heart health? Study finds citrus compounds influence genes linked to blood vessel function (/news/20260916/Could-a-daily-glass-of-orange-juice-support-heart-health-Study-finds-citrus-compounds-influence-genes-linked-to-blood-vessel-function.aspx)

About (/medical/about)

Functional Food (/condition/Functional-Food)

News (/medical/news)

Health A-Z (/medical-a-z.aspx)

Drugs (/drugs-a-z.aspx)

Medical Devices (/Clinical-and-Diagnostics)

Study defines early-onset cirrhosis caused by fatty liver disease (/news/20260916/Study-defines-early-onset-cirrhosis-caused-by-fatty-liver-disease.aspx)

Interviews (/medical/interviews)

White Papers (/medical/whitepapers)

More...

Salivary cortisol patterns predict cognitive decline in older adults (/news/20260918/Salivary-cortisol-patterns-predict-cognitive-decline-in-older-adults.aspx)

POTS identified in nearly two-thirds of Long COVID patients (/news/20260916/POTS-identified-in-nearly-two-thirds-of-Long-COVID-patients.aspx)

Study links missing brain protein to ADHD-like behaviors (/news/20260916/Study-links-missing-brain-protein-to-ADHD-like-behaviors.aspx)

Latest News

Functional Food (/condition/Functional-Food)

Mpox study reveals high rate of false-positive PCR tests (/news/20260922/Mpox-study-reveals-high-rate-of-false-positive-PCR-tests.aspx) News (/medical/news)

Health A-Z (/medical-a-z.aspx)

Drugs (/drugs-a-z.aspx)

Medical Devices (/Clinical-and-Diagnostics)

Interviews (/medical/interviews)

Retinal dopamine levels rise during inherited retinal degeneration (/news/20260922/Retinal-dopamine-levels-rise-during-inherited-retinal-degeneration.aspx) White Papers (/medical/whitepapers)

More...

(/news/20260922/Retinal-dopamine-levels-rise-during-inherited-retinal-degeneration.aspx)

Poor sleep health in late pregnancy linked to higher childhood obesity risk (/news/20260922/Poor-sleep-health-in-late-pregnancy-linked-to-higher-childhood-obesity-risk.aspx)

(/news/20260922/Poor-sleep-health-in-late-pregnancy-linked-to-higher-childhood-obesity-risk.aspx)

Garvan research reveals hidden inflammation behind high colon cancer risk (/news/20260922/Garvan-research-reveals-hidden-inflammation-behind-high-colon-cancer-risk.aspx)

(/news/20260922/Garvan-research-reveals-hidden-inflammation-behind-high-colon-cancer-risk.aspx)

(/)

About (/medical/about)

(/news/20260922/City-of-Hope-develops-adaptive-plug-and-play-CAR-T-technology-to-attack-tumors.aspx)

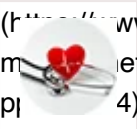
Functional Food (/condition/Functional-Food)

News (/medical/news)

Health A-Z (/medical-a-z.aspx)

Newsletters you may be interested in

Drugs (/drugs-a-z.aspx)

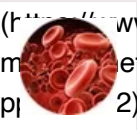


Cardiology (<https://www.news-medical.net/newsletters/webview/?ppnid=154>)
(Subscribe or Preview (<https://www.news-medical.net/newsletters/webview/?ppnid=154>))

Medical Devices (/Clinical-and-Diagnostics)

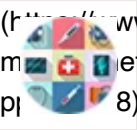
Interviews (/medical/interviews)

White Papers (/medical/whitepapers)



Hematology (<https://www.news-medical.net/newsletters/webview/?ppnid=152>)
(Subscribe or Preview (<https://www.news-medical.net/newsletters/webview/?ppnid=152>))

More...



Medical Device (<https://www.news-medical.net/newsletters/webview/?ppnid=168>)
(Subscribe or Preview (<https://www.news-medical.net/newsletters/webview/?ppnid=168>))

See all Newsletters » (/medical/newsletters)

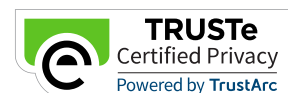


High Credibility: This website adheres to all nine standards of credibility and transparency.



Patient Information Forum

(<https://piftick.org.uk/about-pif-tick/>)



(<https://privacy.truste.com/privacy-seal/validation?rid=80814776-9892-4bc0-a905-50180d9d0296>)



([https://registry.blockmarktech.com/certificates/43640686-6096-4b04-9aee-](https://registry.blockmarktech.com/certificates/43640686-6096-4b04-9aee-5b8e5cf89630/)

5b8e5cf89630/)

Medical Links

[Medical Home \(/medical\)](/medical/)
[Functional Food \(/condition/Functional-Food\)](/condition/Functional-Food/)
[News \(/medical/news\)](/medical/news/)
[Health A-Z \(/medical-a-z.aspx\)](/medical-a-z.aspx)
[White Papers \(/medical/whitepapers\)](/medical/whitepapers/)
[Thought Leaders \(/medical/thought-leaders\)](/medical/thought-leaders/)
[Insights \(/medical/insights-from-industry\)](/medical/insights-from-industry/)
[MediKnowledge Series \(/mediknowledge\)](/mediknowledge/)
[Health & Personal Care \(/Consumer-Products\)](/Consumer-Products/)
[Medical Devices \(/Clinical-and-Diagnostics\)](/Clinical-and-Diagnostics/)
[Drugs \(/drugs-a-z.aspx\)](/drugs-a-z.aspx)
[eBooks \(/medical/ebooks\)](/medical/ebooks/)
[Posters \(/medical/posters\)](/medical/posters/)
[Podcasts \(/medical/podcasts\)](/medical/podcasts/)
[Videos \(/medical/videos\)](/medical/videos/)

Life Sciences Links

[Functional Food \(/condition/Functional-Food\)](/condition/Functional-Food/)
[News \(/life-sciences/news\)](/life-sciences/news/)
[News \(/medical/news\)](/medical/news/)
[Lab Instruments & Equipment \(/Life-Science-and-Laboratory\)](/Life-Science-and-Laboratory/)
[Health A-Z \(/medical-a-z.aspx\)](/medical-a-z.aspx)
[Life Sciences A-Z \(/life-sciences-a-z\)](/life-sciences-a-z/)
[White Papers \(/life-sciences/whitepapers\)](/life-sciences/whitepapers/)
[Drugs \(/drugs-a-z.aspx\)](/drugs-a-z.aspx)
[Thought Leaders \(/life-sciences/thought-leaders\)](/life-sciences/thought-leaders/)
[Insights \(/life-sciences/insights-from-industry\)](/life-sciences/insights-from-industry/)
[Webinars \(/life-sciences/webinars\)](/life-sciences/webinars/)
[Interviews \(/medical/interviews\)](/medical/interviews/)
[eBooks \(/life-sciences/ebooks\)](/life-sciences/ebooks/)
[Posters \(/life-sciences/posters\)](/life-sciences/posters/)
[White Papers \(/medical/whitepapers\)](/medical/whitepapers/)
[Podcasts \(/life-sciences/podcasts\)](/life-sciences/podcasts/)
[Videos \(/life-sciences/videos\)](/life-sciences/videos/)
[More...](#)

Other Useful Links

[About \(/medical/about\)](/medical/about/)
[Meet the Team \(/medical/team\)](/medical/team/)
[Search \(/medical/search\)](/medical/search/)
[Newsletters \(/medical/newsletters\)](/medical/newsletters/)
[Sitemap \(/sitemap.axd\)](/sitemap.axd)
[Advertise \(/medical/advertise\)](/medical/advertise/)
[Contact \(/medical/contact\)](/medical/contact/)
[Terms & Conditions \(/medical/terms\)](/medical/terms/)
[Privacy & Cookie Policy \(/medical/privacy\)](/medical/privacy/)
[Content Policy \(/medical/content-policy\)](/medical/content-policy/)
[Ownership & Funding Information \(/medical/about#ownership-funding\)](/medical/about#ownership-funding)

Other AZoNetwork Sites

[AZoM \(https://www.azom.com/\)](https://www.azom.com/)
[AZoNano \(https://www.azonano.com/\)](https://www.azonano.com/)
[AZoCleantech \(https://www.azocleantech.com/\)](https://www.azocleantech.com/)
[AZoOptics \(https://www.azooptics.com/\)](https://www.azooptics.com/)
[AZoRobotics \(https://www.azorobotics.com/\)](https://www.azorobotics.com/)
[AZoSensors \(https://www.azosensors.com/\)](https://www.azosensors.com/)
[AZoMining \(https://www.azomining.com/\)](https://www.azomining.com/)
[AZoQuantum \(https://www.azoquantum.com/\)](https://www.azoquantum.com/)
[AZoBuild \(https://www.azobuild.com/\)](https://www.azobuild.com/)
[AZoLifeSciences \(https://www.azolifesciences.com/\)](https://www.azolifesciences.com/)



About (/medical/about)

News-Medical.Net provides this medical information service in accordance with these terms and conditions (/medical/terms). Please note that medical information found on this website is designed to support, not to replace the relationship between patient and physician/doctor and the medical advice they may provide.

News (/medical/news)

in



([https://www.linkedin.com/company/news-](https://www.linkedin.com/company/news-medical)

Update Your Privacy Preferences

(<https://www.linkedin.com/company/news-medical>)

Health A-Z (/medical-a-z.aspx)

Last Updated: Tuesday 22 Sep 2026



Drugs (/drugs-a-z.aspx)
(<https://www.azonetwork.com/>)

News-Medical.Net As AZoNetwork Site
News-Medical.Net As AZoNetwork Site

Owned and operated by AZoNetwork, © 2000-2026

Interviews (/medical/interviews)

White Papers (/medical/whitepapers)

More...